

ANNUAL RAINFALL PATTERNS ACROSS ADELAIDE, PERTH AND SYDNEY, 1976-2025

Group Griffin

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ABSTRACT

This report compares annual rainfall in Adelaide, Perth and Sydney using 50 years of rainfall records from 1976 to 2025. The aim was to identify how average rainfall and year-to-year variation differ between the three cities. Rainfall data were obtained from the Australian Bureau of Meteorology Climate Data Online service and cleaned before analysis. The study used descriptive statistics, graphs and a Kruskal-Wallis test. Sydney had the highest mean annual rainfall and the largest variation. Perth had moderate rainfall and moderate variation. Adelaide had the lowest mean rainfall and the smallest spread. The Kruskal-Wallis test showed that the annual rainfall distributions were significantly different across the three cities. These findings show that rainfall planning should be based on local climate behaviour rather than broad national averages. This matters for water supply planning, stormwater design and climate adaptation. Future research should include more cities, longer records and seasonal rainfall analysis.

Key words: Annual rainfall; rainfall variability; Kruskal–Wallis test; Australian climate; water planning

INTRODUCTION

Rainfall is a key input for water supply planning, drainage design and drought management. Australia is not climatically uniform, so city-scale analysis is more useful than a single national summary. The Bureau of Meteorology provides historical rainfall records through Climate Data Online, including monthly and annual observations for individual stations (Bureau of Meteorology 2026).

Studies of eastern and southern Australia also show that climate drivers such as the El Nino-Southern Oscillation, Indian Ocean Dipole and Southern Annular Mode can influence rainfall differently across regions (Pepler et al. 2014; Montazerolghaem et al. 2016). These studies suggest that rainfall analysis should consider both the centre of the data and the spread, not only a single average value.

This study compares annual rainfall in Adelaide, Perth and Sydney. The aim is to determine whether annual rainfall patterns differ between the three cities, with specific attention to average rainfall, variability and long-term visual patterns. The project does not attempt to prove climate change at these stations. It is an engineering data analytics study focused on clear comparison and appropriate statistical method selection.

STUDY AREA AND DATA SET

Rainfall data were collected from the Bureau of Meteorology Climate Data Online service. Three station files were used: Adelaide station 023034, Perth station 009021 and Sydney station 066052. Each file contained monthly rainfall and an annual rainfall total in millimetres. The analysis used annual rainfall from 1976 to 2025, giving 50 observations for each city and 150 observations in total. The files were imported into R as comma-separated value files. The data cleaning process kept the Year and Annual rainfall fields, converted Annual rainfall to numeric format, added a City label to each file, removed missing annual values and filtered the records to 1976-2025. After filtering, all three cities retained 50 valid annual values. The study area is shown in Figure 1.

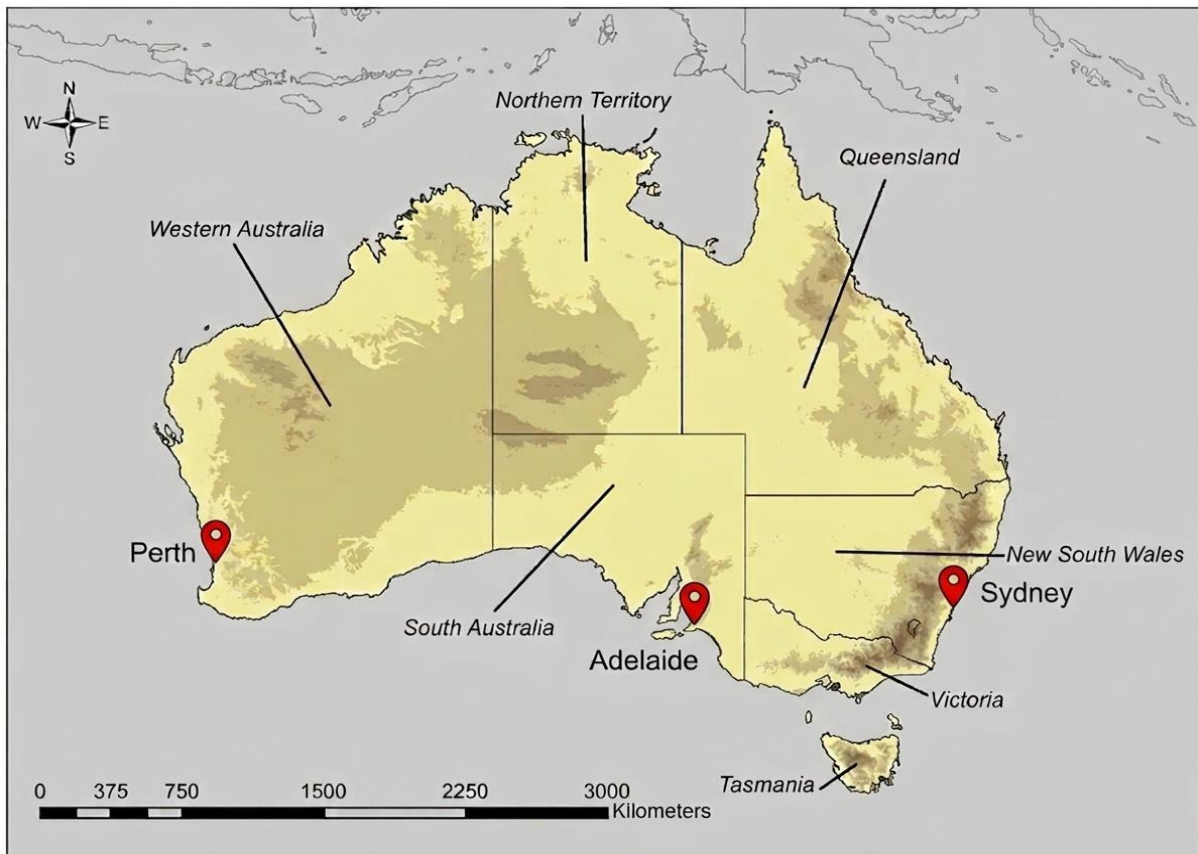


Fig 1 : Location of selected Australian capital cities used in the study.

METHODS

The analysis was completed in R using base statistical functions and ggplot2 for visualisation. First, descriptive statistics were calculated for each city. The table was ordered from minimum to maximum, with standard deviation shown at the end, so the summary is easy to read. Bar charts, boxplots and time-series plots were then used to compare average rainfall, spread, outliers and changes through time.

Several tests were considered and rejected. A z-test was not appropriate because the population variance was not known and there was no fixed population mean being tested. Pairwise t-tests

were not used because the main research question compared three cities at once. A one-way ANOVA was considered, but it was not used as the final test because the rainfall data showed unequal spread and visible outliers. The largest standard deviation was about 3.59 times the smallest, which made the equal-variance assumption doubtful. Instead, the Kruskal-Wallis test was selected because it compares three independent groups using ranked data and is suitable when standard ANOVA assumptions are not reasonable.

The null hypothesis was that the annual rainfall distributions were the same for Adelaide, Perth and Sydney. The alternative hypothesis was that at least one city had a different annual rainfall distribution. A significance level of 0.05 was used. The workflow is shown in Figure 2.

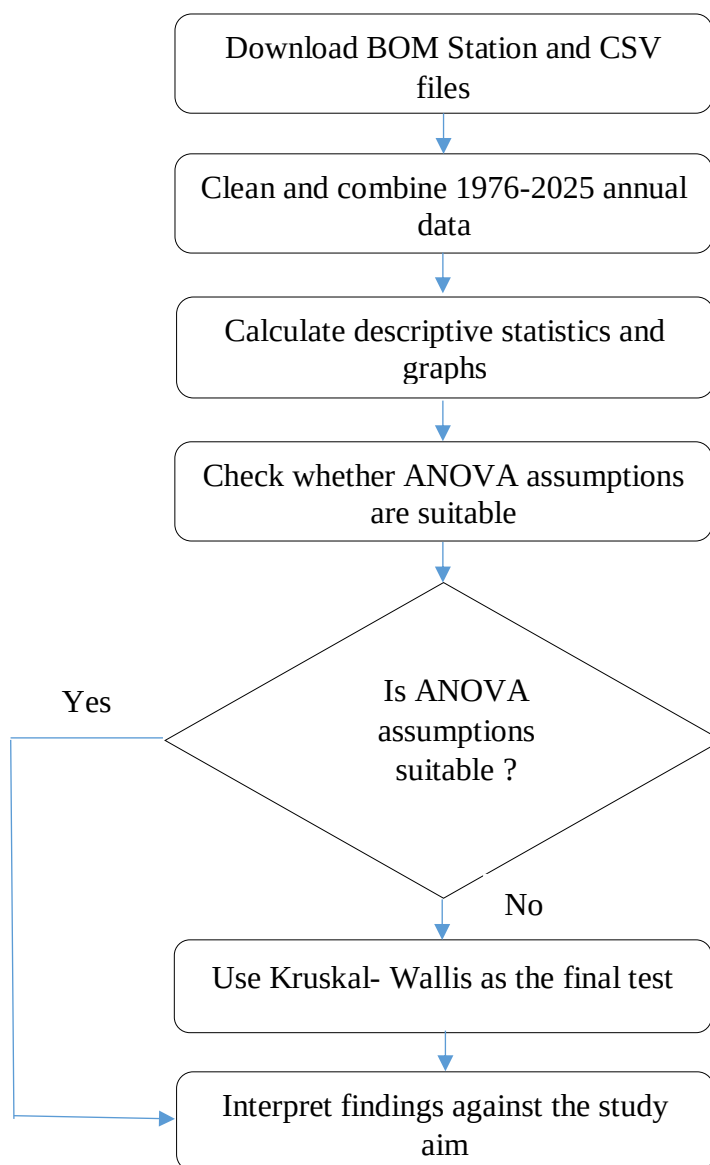


Figure 2. Methodology flow chart.

RESULTS

The descriptive statistics show a clear rainfall gradient across the three cities. Sydney had the highest mean annual rainfall at 1295.23 mm, followed by Perth at 706.68 mm and Adelaide at 426.34 mm. Sydney also had the largest spread, with a standard deviation of 343.07 mm and a maximum annual rainfall of 2620.00 mm. Adelaide had the smallest spread, with a standard deviation of 95.43 mm. This means Sydney was not only wetter, but also less consistent from year to year.

City	Min	Q1	Median	Mean	Q3	Max	SD
Adelaide	234.60	376.10	422.96	426.34	480.90	730.80	95.43
Perth	479.60	621.35	696.35	706.68	793.15	960.00	111.60
Sydney	706.90	1085.10	1297.50	1295.23	1470.88	2620.00	343.07

Table 1. Descriptive statistics for annual rainfall, 1976-2025.

Figure 3 confirms the same ranking in a simpler visual form. One colour is used because the city names are already identified on the x-axis. Figure 4 shows that Adelaide values are concentrated at the lower end, Perth values are higher but still fairly compact, and Sydney values are much higher with a wider spread and a high outlier. Figure 5 shows strong year-to-year fluctuations in all three cities. The same line type is used for all cities, with colour used only to separate the city series.

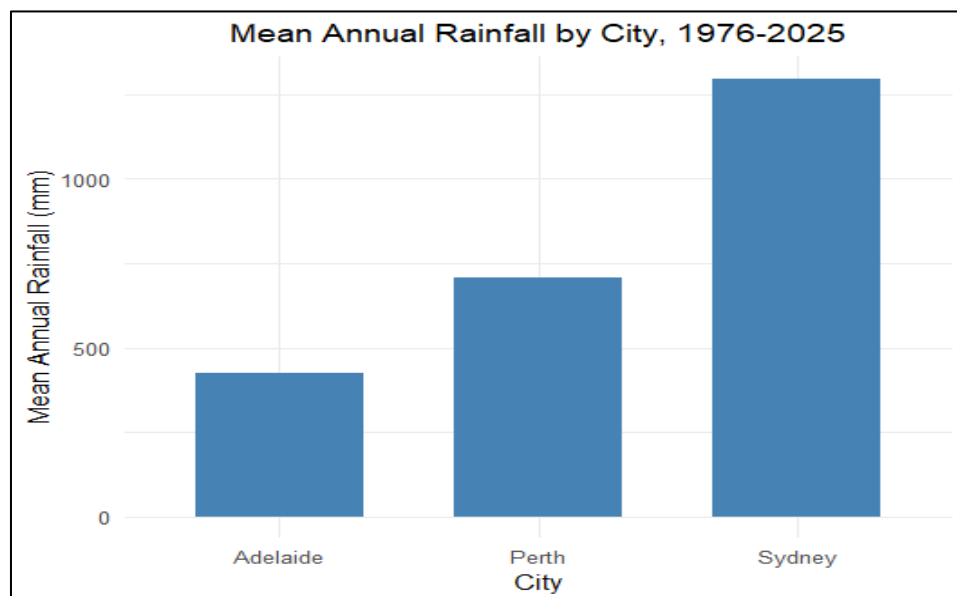


Figure 3. Bar chart for mean annual rainfall, 1976-2025.

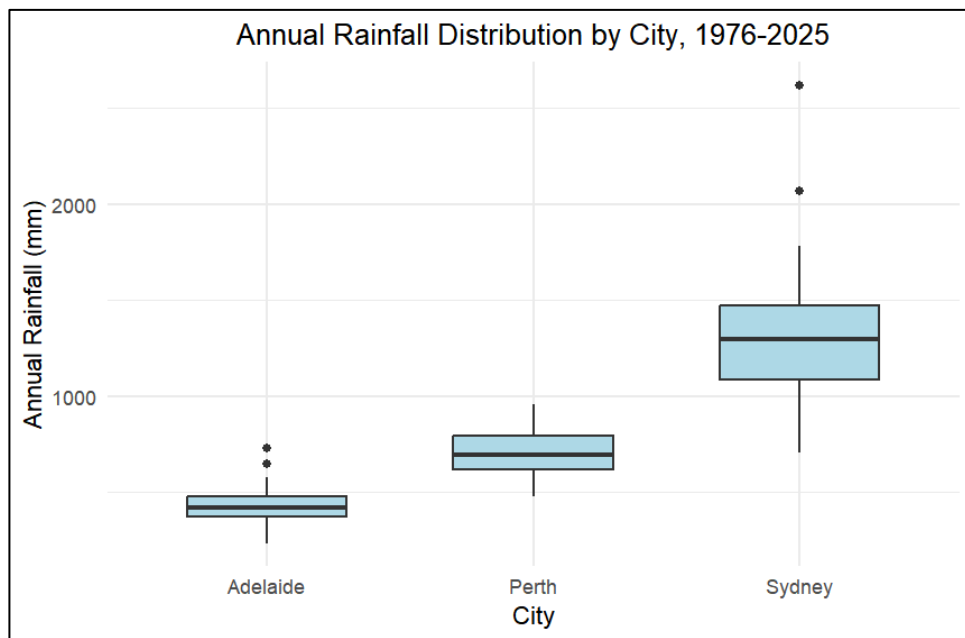


Figure 4. Box plot for mean annual rainfall, 1976-2025.

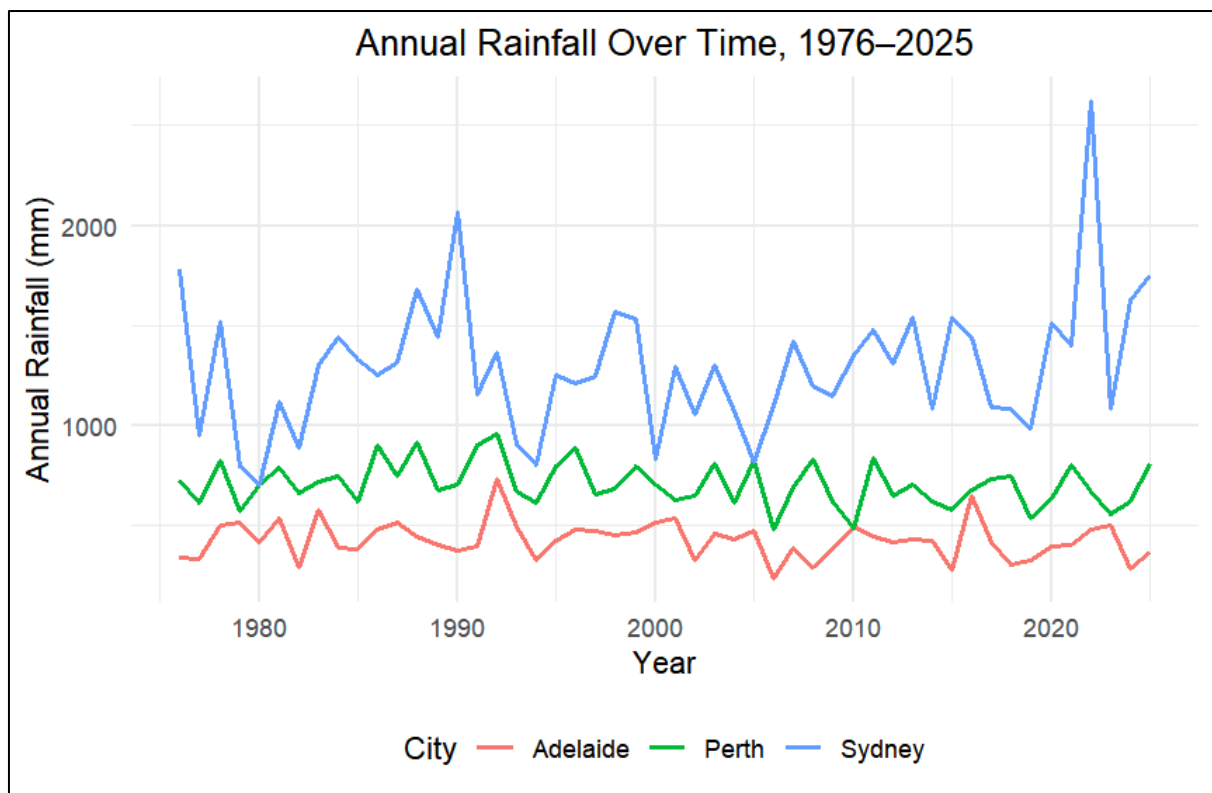


Figure 5. Line graph for mean annual rainfall, 1976-2025.

The Kruskal-Wallis test gave $H = 124.88$, with 2 degrees of freedom and $p < 0.001$. Since the p-value was below 0.05, the null hypothesis was rejected. This result provides strong statistical evidence that the annual rainfall distributions are not the same across Adelaide, Perth and Sydney. The statistical result supports the descriptive findings and directly answers the project aim.

CONCLUSION

This project found clear differences in annual rainfall patterns across Adelaide, Perth and Sydney. Sydney was the wettest city and had the greatest year-to-year variability. Perth had moderate annual rainfall. Adelaide was the driest city and had the most stable annual rainfall. The Kruskal-Wallis test confirmed that the annual rainfall distributions differed significantly across the three cities. Therefore, the project aim was achieved: average rainfall and variability were not the same for the three cities.

The findings have practical implications. Water planning, stormwater design and drought preparation should be based on local rainfall behaviour rather than broad national averages. Sydney may need greater attention to high-rainfall years and stormwater capacity, while Adelaide needs careful long-term water security planning because of its lower rainfall. Perth sits between these cases but still shows a rainfall pattern that is different from Adelaide and Sydney.

Future research should include more Australian cities, longer station records and seasonal rainfall analysis. It should also include climate drivers such as the El Nino-Southern Oscillation, Indian Ocean Dipole and Southern Annular Mode. These additions would help explain why rainfall changes between years, not only whether rainfall differs between cities.

BIBLIOGRAPHY

Bureau of Meteorology 2026, Climate Data Online, Australian Government, viewed 28 May 2026, <<https://www.bom.gov.au/climate/data/>>.

Helsel, DR, Hirsch, RM, Ryberg, KR, Archfield, SA & Gilroy, EJ 2020, *Statistical methods in water resources*, Techniques and Methods 4-A3, US Geological Survey, viewed 28 May 2026, <https://doi.org/10.3133/tm4a3>

Pepler, A, Timbal, B, Rakich, C & Coutts-Smith, A 2014, 'Indian Ocean Dipole overrides ENSO's influence on cool season rainfall across the eastern seaboard of Australia', *Journal of Climate*,, <https://journals.ametsoc.org/view/journals/clim/27/10/jcli-d-13-00554.1.xml>.

Posit Software, PBC 2026, RStudio IDE User Guide

GITHUB REPOSITORY

<https://github.com/pantasanish4-svg/Rainfall-patterns-comparison>